Nicholas Tagliamonte

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Summary

Software developer and systems researcher working across enterprise applications, information retrieval, and reliable distributed storage. Experience designing and implementing distributed infrastructure in Go and production software spanning document management, hybrid search, and geometric PDF analysis.

Education

2028 M.S., Computer Science, **University of Pennsylvania** (Admitted)
2026 B.S. with Distinction, Computer Science, Mathematics Minor, **Temple University** (GPA: 3.9/4.0)
2024 A.S., Computer Science; A.S., Mathematics, **Bucks County CC** (GPA: 4.0/4.0)

Research and Publications

Tarsus: Latency-Optimal Policy-Decoupled Distributed Storage
First Author; Accepted at ICICT 2027

Designed and implemented a policy-decoupled distributed storage system and developed its experimental evaluation. The architecture separates changing reliability policy from the data path while targeting latency-optimal operation. Co-authored with Justin Y. Shi and Garrett Stoll.

Probabilistic Decentralized Storage via Provider-Data Decoupling
Patent Pending; Co-Inventor, 2026

Co-invented a distributed file system architecture based on IPFS with an enforced replication model. Contributed core system design and theoretical foundations.

Work Experience

Software Developer – AI/ML, R&D - iPipeline Jun 2026 - Present

- Designed, proposed, and implemented a document-versioning system backed by a Merkle DAG.
- Researched and integrated BM25 into an existing vector-retrieval pipeline, creating hybrid search that improved retrieval reliability.
- Implemented a geometric input-field detection algorithm that increased bounding-box placement accuracy from 34% to 92.66%.
- Develop and maintain enterprise application features and services; contribute to technical design, code review, reliability, and performance work.

Teaching Assistant - CIS 3308, Temple University Aug 2025 - May 2026

- Led labs and office hours for approx. 70 students for 2 semesters covering React, SQL, and Spring Boot.
- Helped students debug apps, understand software design concepts, and improve development workflows.
- Nominated for Temple University Outstanding Undergraduate Teaching Assistant Award (2026).

Software Developer Intern - iPipeline Jun 2025 - Aug 2025

- Migrated CI/CD pipelines for 4 client-facing development teams from Jenkins to GitHub Actions, standardizing deployment workflows and reducing maintenance overhead across projects.
- Created PowerShell automation tools reducing UI test cycles by ~ 90 s per screen; adopted company-wide.
- Implemented front-end features, PDF mapping logic, and XML integrations for client-facing systems.
- Supported AWS serverless migration (Terraform/Terragrunt configuration and unit tests).

IT Service Desk Regional Lead - Victrex Dec 2023 - Sep 2024

– Led regional IT operations supporting 2,000 employees across North America, managing $60k operational budget.
– Directed network cutover project and coordinated vendor operations across multiple vendors.

Global IT Helpdesk Analyst - Victrex Jul 2022 - Jan 2024

– Provided first-line IT support with >70% first-call resolution; maintained internal knowledge base and supported Azure AD environments.

Projects

Managed AI Infrastructure Platform *Link*

Architected a managed RAG API using FastAPI, vLLM, and Pgvector on GCP to provide centralized vector search and LLM capabilities for multiple student development teams. Engineered asynchronous ingestion pipeline for document processing, reducing upload latency by $\sim 20\times$ and eliminating timeout failures for files >1 MiB.

Parallel Distributed Matrix Multiplication *Link*

A fault-tolerant parallel matrix multiplication framework in C using a custom tuple space server (TSH) for distributed coordination. Includes fault-tolerant execution under worker failure, dynamic worker spawning, tunable task granularity, and benchmarking scripts.

Number Theory Calculator *Link*

Originally designed as a GUI replica of the TI-30X IIS, expanded into a cross-platform scientific and number-theoretic calculator. Implements modular arithmetic, factorization, totient and Carmichael functions, generator and residue tests, discrete log estimation, and related tools. Features customizable insert mode, improved statistical input, detailed error handling, and platform-specific save/load logic.

Skills

Languages	Go, C, Python, Java, JavaScript, SQL, HTML/CSS
Frameworks & Libraries	React, FastAPI, Terraform, GitHub Actions, libp2p, Boxo, Bitswap
Cloud & Infrastructure	AWS, GCP, Azure, CI/CD, Docker, Git
Tools	PowerShell, Linux, LaTeX